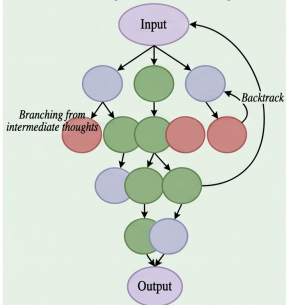


Tree of Thoughts

Branching search with backtracking

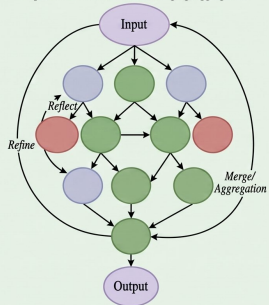


Key novelty:
Generating multiple thoughts from any node, exploring and backtracking through the tree.

Legend: ● Unscores ● High score ● Low score

Graph of Thoughts

Graph search with refinement, merging, aggregation

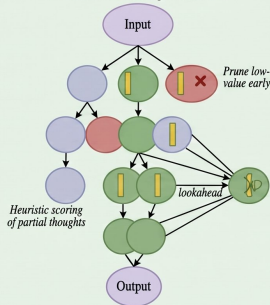


Key novelty (beyond ToT):
Arbitrary graph transformations: aggregation, refinement loops, and merging of chains.

→ Expansion -- → Backtrack/lookahead - Prune

Heuristic Tree of Thoughts

ToT with heuristic scoring and lookahead

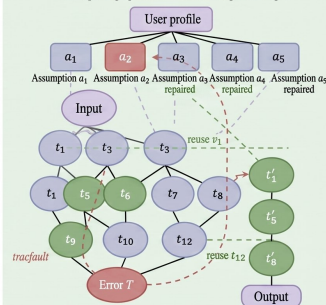


Key novelty (beyond ToT):
Heuristic scoring $h(t)$ with lookahead evaluation enables early pruning of low-value subtrees.

■ Assumption - - - Posterior attr.

Intro-Specter

Assumption-graph abduction and user-ground repair



Backward attribution to assumptions Update $p(a_h | \text{error})$ Repair only affected steps, reuse others

Key novelty (beyond GoT/ToTH):
Abductive inference over a probabilistic assumption graph; posterior-driven fault attribution through the reasoning DAG, repairing only the affected subgraph while reusing valid steps.

Figure 2: Comparison of Tree of Thoughts (ToT), Graph of Thoughts (GoT), Heuristic Tree of Thoughts (ToTH), and Intro-Specter.

→ dependency edge

- - - ➔ counterfactual attribution